

Kotlin Coverage Baseline —

2026-06-04

Phase-0 / §6.D behavioral-coverage baseline for the Kotlin (Android client + api-app) tree, captured by **Kover 0.9.1** wired through the `StaticAnalysisConventionPlugin buildSrc` convention (applied to every module). Later completeness-program phases drive these numbers up; this file is the starting point against which progress is measured.

How it was produced

```
nice -n 19 ./gradlew --no-daemon --max-workers=2 --continue \
  koverXmlReport koverHtmlReport \
  -x :feature:credentials:testDebugUnitTest \
  -x :feature:credentials:testReleaseUnitTest
```

- Real test execution → real coverage (Kover instruments the actual unit-test run).
- The two excluded `:feature:credentials` test tasks carry the **pre-existing flaky** test `CredentialsViewModelTest > select provider updates selectedProvider (TurbineAssertionError: No value produced in 3s)` — the same timing flake the constitution's §6.L 68th-cycle note records as already-known. It is unrelated to this build-wiring change (which touches only `buildSrc/`, `gradle/`, `config/detekt/`). Excluding it lets the full tree's coverage report complete; `core/credentials` coverage below reflects its non-flaky tests that still ran.
- Raw per-module Kover XML reports archived at `.lava-ci-evidence/completeness-program/coverage/kotlin-baseline-2026-06-04/`.

Headline (aggregate across all 55 modules)

Counter	Coverage	Covered / Total
LINE	18.65%	4709 / 25244
BRANCH	20.01%	1692 / 8457

Counter	Coverage	Covered / Total
INSTRUCTION	15.91%	35470 / 222985
METHOD	17.14%	1189 / 6939
CLASS	16.91%	508 / 3004

No root Kover aggregation project exists (there is no root `build.gradle.kts`, per the convention-plugin architecture), so these aggregate figures are summed from the 55 per-module `report.xml` counters. Per-module `koverHtmlReport` output lives under each module's `build/reports/kover/html/`.

Per-module line + branch coverage

Module	Line %	Line cov/total	Branch %	Branch cov/total
core/tracker/gutenberg	95.41%	208/218	51.33%	77/150
core/tracker/archiveorg	90.91%	190/209	47.31%	88/186
core/tracker/mirror	88.89%	8/9	50.00%	2/4
core/tracker/rutor	83.50%	496/594	56.84%	245/431
core/apiengine	83.46%	106/127	36.36%	32/88
core/tracker/kinozal	83.16%	237/285	41.18%	56/136
core/tracker/nmclub	82.34%	289/351	47.06%	104/221
core/tracker/registry	75.00%	3/4	0.00%	0/2
core/sync	65.22%	15/23	0.00%	0/0
core/credentials	58.46%	235/402	59.68%	74/124
core/tracker/client	49.82%	410/823	52.38%	110/210
core/tracker/rutracker	43.53%	632/1452	30.14%	211/700
core/auth/impl	40.62%	26/64	25.00%	6/24
core/network/impl	37.43%	189/505	42.86%	84/196
core/tracker/api	37.21%	48/129	15.60%	34/218
feature/onboarding	30.22%	243/804	30.75%	103/335
feature/login	28.76%	266/925	30.28%	86/284
core/tracker/testing	28.57%	24/84	8.33%	3/36

Module	Line %	Line cov/ total	Branch %	Branch cov/ total
core/testing	26.77%	34/127	75.00%	6/8
api-app	19.50%	118/605	19.28%	32/166
feature/credentials	19.32%	96/497	16.55%	24/145
feature/menu	18.27%	129/706	15.48%	39/252
feature/connection	16.07%	58/361	20.00%	22/110
feature/ provider_config	15.00%	81/540	8.99%	16/178
core/preferences	13.85%	27/195	22.67%	17/75
core/network/api	13.71%	41/299	4.39%	15/342
feature/search_result	11.04%	146/1323	8.02%	49/611
app	9.45%	52/550	24.10%	40/166
core/models	9.22%	19/206	71.43%	30/42
core/data	6.84%	62/906	10.30%	41/398
core/domain	6.75%	86/1274	15.00%	24/160
core/designsystem	5.62%	135/2400	0.85%	4/472
core/auth/api	0.00%	0/4	0.00%	0/0
core/common	0.00%	0/8	0.00%	0/0
core/database	0.00%	0/3187	0.00%	0/428
core/dispatchers	0.00%	0/11	0.00%	0/0
core/downloads	0.00%	0/97	0.00%	0/59
core/logger	0.00%	0/23	0.00%	0/0
core/navigation	0.00%	0/309	1.09%	1/92
core/notifications	0.00%	0/77	0.00%	0/6
core/ui	0.00%	0/862	0.33%	1/307
core/work/api	0.00%	0/0	0.00%	0/0
core/work/impl	0.00%	0/270	0.00%	0/48
feature/account	0.00%	0/115	3.85%	1/26
feature/bookmarks	0.00%	0/159	0.00%	0/38
feature/category	0.00%	0/297	3.77%	4/106
feature/ credentials_manager	0.00%	0/392	0.75%	1/134
feature/favorites	0.00%	0/124	2.50%	1/40
feature/forum	0.00%	0/229	0.00%	0/46

Module	Line %	Line cov/ total	Branch %	Branch cov/ total
feature/main	0.00%	0/42	0.00%	0/28
feature/rating	0.00%	0/156	2.22%	1/45
feature/search	0.00%	0/425	0.71%	1/140
feature/search_input	0.00%	0/362	0.00%	0/148
feature/topic	0.00%	0/1015	1.88%	5/266
feature/visited	0.00%	0/83	6.67%	2/30

Reading the baseline

- **Best-covered:** the tracker SDK plugin modules — core/tracker/gutenberg (95.41%), core/tracker/archiveorg (90.91%), core/tracker/rutor (83.50%), core/tracker/kinozal (83.16%), core/tracker/nmclub (82.34%), and core/apiengine (83.46%). These are the §6.D behaviorally-tested surfaces.
- **0% modules** are dominated by (a) generated code with no behavioral tests (core/database = 3187 lines of Room-generated DAOs/schema, core/ui, core/navigation) and (b) Compose feature screens whose only coverage path is the androidTest Challenge suite (NOT counted here — Kover's koverXmlReport measures JVM unit tests only; instrumentation coverage is a separate, later measurement). feature/topic, feature/search, feature/category etc. show 0% because their behavior is exercised by Compose UI Challenge Tests on the emulator matrix, not by JVM unit tests.
- **Drivers of the low aggregate:** core/designsystem (2400 lines, 5.62%), core/database (3187 lines, 0%), feature/search_result (1323 lines, 11%), core/domain (1274 lines, 6.75%), and feature/topic (1015 lines, 0%) are the largest uncovered surfaces and the highest-leverage targets for later phases.

Phase targets (informational — set by later completeness-program phases)

The completeness program drives LINE coverage up from this 18.65% baseline. The highest-leverage, lowest-risk targets (large + behaviorally-testable at the JVM unit level, not Compose-only):

1. core/domain use cases (1274 lines, 6.75%) — pure Kotlin, ViewModel-reachable.

2. core/data repositories (906 lines, 6.84%) — fakes already exist in core:testing.
3. core/network/impl (505 lines, 37%) and core/tracker/client (823 lines, 50%).
4. feature/* ViewModels via orbit-test (wired in every feature, currently unused).